A Face-Projection Spin(10) Effective Framework: Rubik Covariance, $O(2)$ Family Geometry, and the Conditional GUT Boundary

Working draft

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Abstract

This draft rewrites the theory around a face-projection principle. Low-energy Standard-Model labels are not treated as primitive entries in a table; they are the projections of a higher-dimensional internal representation under a chosen symmetry-breaking face. In this language, one Spin(10) half-spinor 16 is the standard minimal six-face realization whose Standard-Model projection displays

$$Q, \quad L, \quad u^c, \quad d^c, \quad \nu^c, \quad e^c.$$

Three families are protected separately by the index

$$H^0(\mathbb{CP}^1, \mathcal{O}(2)), \qquad \dim H^0(\mathbb{CP}^1, \mathcal{O}(2)) = 3.$$

The result is not an unconditional PSLT-only derivation of a unique grand unified theory. It is a conditional Spin(10) effective framework. The Majorana contact tensor direction is no longer left arbitrary: it is the unique $SL(2)$-invariant second-transvectant direction K_{tr}. An optional Route-B hidden-messenger mechanism for ζK_{tr} is recorded as a conditional extension, but it is not part of the Route-A theorem core. The hidden phase/radial sector and the companion audits for flavor, proton decay, thresholds, and source-sector UV origin remain open. Detailed machine-audit and scan appendices have been removed from this version until it is clear which of them truly belong to the face-projection framework.

1 Scope of the Lean Route-A Note

This version is deliberately a Route-A paper skeleton. Its theorem-level claims are limited to three representation-geometric statements:

six faces	Spin(10) half-spinor restriction gives one SM family plus ν^c,
three copies	$\mathrm{Spin}^c(\mathbb{CP}^1, \mathcal{O}(2))$ Dolbeault index gives three protected sections,
contact direction	K_{tr} is the unique $SL(2)$ second-transvectant direction.

Route B is kept only as a conditional candidate mechanism for the complex coefficient ζ. Full CKM/PMNS fitting, dimension-five proton-decay Wilson tensors, threshold matching, and source-sector UV completion are not claimed in this lean note; they remain companion-audit problems. Thus the paper should be read as a conditional face-projection Spin(10) EFT note, not as a completed first-principles derivation of a unique GUT.

2 Face Projection

The organizing idea is that the visible particle table is a projection table, not a table of fundamental labels. Let G be a high-energy compact group, let R be a complex representation of G,

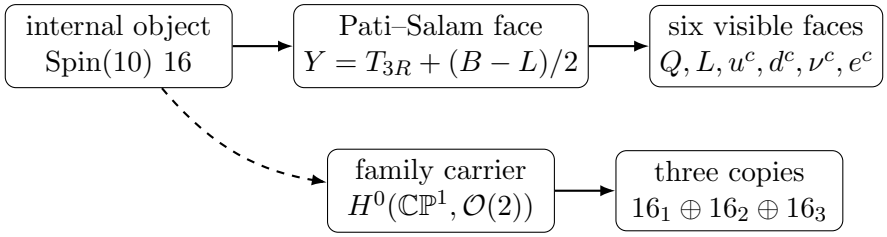

Figure 1: Face-projection picture. The six Standard-Model multiplet types are visible faces of one Spin(10) half-spinor, while the family number is an independent protected index.

and let a vacuum orientation θ select a broken subgroup

$$H_\theta \simeq \mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y.$$

The face projection is the restriction functor

$$\Pi_\theta(R) := \mathrm{Res}^G_{H_\theta} R = \bigoplus_\alpha m_{\alpha,\theta}\, r_{\alpha,\theta},$$

where the irreducible H_θ-summands $r_{\alpha,\theta}$ are the visible multiplet types. If $g \in G$, then the embedded subgroup changes by conjugation,

$$H_\theta \longmapsto H_{g\theta} = g H_\theta g^{-1}.$$

The same high-energy representation can therefore display different visible labels under a different face:

$$\Pi_{g\theta}(R) = \mathrm{Res}^G_{gH_\theta g^{-1}} R.$$

This is the representation-theoretic version of the intuitive statement that the same internal object can look different after a legal internal rotation.

Definition 2.1 (Rubik covariance)**.** A unified model is face-covariant if the visible low-energy multiplet types are irreducible summands in restrictions of a common high-energy representation, and if every allowed change of face is a conjugation of the broken-subgroup embedding preserving anomaly cancellation, index, chirality, charge normalization, and representation dimension. The principle does not allow arbitrary transmutation of low-energy particles. It says that different low-energy labels can be different visible faces of the same high-energy object.

The Rubik-cube image is useful only in this precise sense. A chemical element is still defined by nuclear charge; it does not become another element because one rotates a drawing. The analogy points instead to root lattices, Weyl chambers, gauge-basis rotations, and symmetry-breaking projections.

The concrete grand-unification branch used below follows the standard Pati–Salam and Spin(10) route [1, 2, 3, 4]. The conditional neutrino sector uses the usual type-I seesaw logic [5, 6, 7], while the threshold language is the conventional perturbative RGE language [8, 9].

3 The Six-Face Spin(10) Object

Proposition 3.1 (Standard minimal six-face realization)**.** *Among standard simple compact GUT candidates, if the six chiral Standard-Model multiplet types*

$$Q, \quad L, \quad u^c, \quad d^c, \quad \nu^c, \quad e^c$$

are required to arise from one irreducible complex chiral representation whose restriction contains exactly one SM family plus ν^c, with no chiral SM exotics and with minimal representation dimension, the standard minimal single-object realization is the Spin(10) *half-spinor* 16*.*

Proof. The Pati–Salam subgroup of Spin(10) gives

$$16 \to (4,2,1) \oplus (\bar{4},1,2).$$

Breaking

$$\mathrm{SU}(4)_C \to \mathrm{SU}(3)_C \times \mathrm{U}(1)_{B-L}$$

gives

$$4 \to (3,1/3) \oplus (1,-1), \qquad \bar{4} \to (\bar{3},-1/3) \oplus (1,+1).$$

With

$$Y = T_{3R} + \frac{B-L}{2},$$

the restriction to the Standard-Model face is

$$\mathrm{Res}^{\mathrm{Spin}(10)}_{G_{\mathrm{SM}}}\, 16 = Q \oplus L \oplus u^c \oplus d^c \oplus \nu^c \oplus e^c,$$

where the six projected Standard-Model multiplet types are

field	$\mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$
Q	$(3,2)_{1/6}$
L	$(1,2)_{-1/2}$
u^c	$(\bar{3},1)_{-2/3}$
d^c	$(\bar{3},1)_{1/3}$
ν^c	$(1,1)_0$
e^c	$(1,1)_1$.

The $SU(5)$ alternative uses $10 \oplus \bar{5} \oplus 1$, which is not one irreducible chiral object. Pati–Salam is natural but not simple. E_6 contains this structure inside a larger 27, usually bringing extra matter that must be lifted. Therefore Spin(10) with a 16 is the standard minimal single-object realization. □

Proposition 3.2 (Hypercharge normalization)**.** *The face projection*

$$Y = T_{3R} + \frac{B-L}{2}$$

has the GUT normalization

$$\frac{\mathrm{Tr}\, Y^2}{\mathrm{Tr}\, T_{3L}^2} = \frac{5}{3}.$$

Proof. The trace is over one left-handed 16 after Standard-Model decomposition. For one family,

$$\mathrm{Tr}\, Y^2 = 6\left(\frac{1}{6}\right)^2 + 2\left(-\frac{1}{2}\right)^2 + 3\left(-\frac{2}{3}\right)^2 + 3\left(\frac{1}{3}\right)^2 + 1(1)^2 + 1(0)^2 = \frac{10}{3}.$$

The $SU(2)_L$ doublets are Q and L, and the doublet normalization is $\mathrm{Tr}_{\mathbf{2}}\, T_3^2 = 1/2$. Hence

$$\mathrm{Tr}\, T_{3L}^2 = 3 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = 2.$$

Thus

$$\frac{\mathrm{Tr}\, Y^2}{\mathrm{Tr}\, T_{3L}^2} = \frac{10/3}{2} = \frac{5}{3}.$$

□

Remark 3.3 (What rotation can and cannot mean). At low energy Q cannot be freely rotated into e^c. The Standard-Model gauge symmetry has already selected superselection-like labels. The face-projection statement is about the high-energy representation and its broken-subgroup projection. Any physical process that changes the visible face must be mediated by legitimate high-energy fields and is constrained, for example, by proton decay.

4 Three Families from an $O(2)$ Index

The family number is not produced by rotating the six-face object. It is a separate index statement. The analytic input is the standard Spin^c Dolbeault–Dirac/cohomology identification and the Riemann–Roch–Serre-duality calculation on $\mathbb{CP}^1$ [10, 11].

Theorem 4.1 (Conditional Spin^c index theorem for three protected families)**.** *Assume that the protected internal family carrier is the* Spin^c *Dolbeault bundle on*

$$C = \mathbb{CP}^1$$

twisted by a two-center divisor

$$R = p_+ + p_-, \qquad L_R = \mathcal{O}(R) \simeq \mathcal{O}(2).$$

Assume also that all other SM-charged internal sectors are absent, gapped, or vectorlike, so they contribute no additional unpaired chiral zero modes. Let

$$D_R = \sqrt{2}\left(\bar{\partial}_{L_R} + \bar{\partial}^\dagger_{L_R}\right)$$

be the corresponding Spin^c *Dolbeault–Dirac operator. Then*

$$\ker D_R^+ \simeq H^0(\mathbb{CP}^1, \mathcal{O}(2)), \qquad \ker D_R^- \simeq H^1(\mathbb{CP}^1, \mathcal{O}(2)),$$

with

$$\dim H^0(\mathbb{CP}^1, \mathcal{O}(2)) = 3,$$

and $H^1(\mathbb{CP}^1, \mathcal{O}(2)) = 0$*. Thus the protected sector contributes exactly three positive-chirality family zero modes.*

Proof. Hodge theory identifies the positive and negative Spin^c Dolbeault–Dirac kernels with the Dolbeault cohomology groups displayed above. Riemann–Roch on $\mathbb{CP}^1$ gives

$$h^0(\mathbb{CP}^1, \mathcal{O}(2)) - h^1(\mathbb{CP}^1, \mathcal{O}(2)) = 2 + 1 = 3.$$

Serre duality gives

$$h^1(\mathbb{CP}^1, \mathcal{O}(2)) = h^0(\mathbb{CP}^1, K_{\mathbb{CP}^1} \otimes \mathcal{O}(-2)) = h^0(\mathbb{CP}^1, \mathcal{O}(-4)) = 0.$$

Therefore $h^0 = 3$. □

Corollary 4.2 (Projected chiral state space)**.** *Under the six-face* $\mathrm{Spin}(10)$ *assumption and the* $O(2)$ *family carrier, the low-energy chiral state space is*

$$\mathcal{S}^{4D}_{\mathrm{chiral}} = 16 \otimes H^0(\mathbb{CP}^1, \mathcal{O}(2)) \simeq 16_1 \oplus 16_2 \oplus 16_3.$$

Remark 4.3 (Why this is still conditional). This theorem does not derive $C = \mathbb{CP}^1$, the two-center divisor R, the Spin^c carrier, or the no-extra-unpaired-sector condition. Those are branch assumptions. The theorem says that once those inputs are chosen, the family count is protected and is not a tunable scan output.

5 Conditional EFT Data

The face-projection and index principles supply the representational skeleton. They do not by themselves determine the full action. The current conditional branch retains the following action-level data:

A1 Spin(10) face-projection EFT with complete 16_i,
A2 $H^0(\mathbb{CP}^1, \mathcal{O}(2))$ protects three copies of the 16,
A3 type-I seesaw with a source-fixed Majorana contact,
A4 proton bounds and two-loop threshold matching,
A5 constrained/composite 54/210 source order parameters,
A6 auxiliary/composite conormal multipliers,
A7 a Schur lock with singlet auxiliaries and sterile completion.

Lemma 5.1 (Unique $SL(2)$-invariant contact direction)**.** *Let*

$$V = H^0(\mathbb{CP}^1, \mathcal{O}(2)) \simeq \mathrm{Sym}^2\mathbb{C}^2.$$

Then

$$\mathrm{Sym}^2 V \simeq \mathrm{Sym}^4\mathbb{C}^2 \oplus \mathrm{Sym}^0\mathbb{C}^2.$$

The Veronese moment sector is the $\mathrm{Sym}^4\mathbb{C}^2$ *summand, while the orthogonal singlet contact sector is unique up to phase. In the normalized* (ψ_0, ψ_1, ψ_2) *basis used for the family carrier, this singlet can be represented as*

$$K_{\mathrm{tr}} = \frac{1}{\sqrt{3}} \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}.$$

Coordinate-free, $K_{\mathrm{tr}} \in \mathrm{Sym}^2 V^\vee$ *and its inverse bilinear form is* $K_{\mathrm{tr}}^{-1} \in \mathrm{Sym}^2 V$*. In the displayed orthonormal basis, the induced matrix obeys*

$$K_{\mathrm{tr}}^2 = \frac{1}{3} I, \qquad K_{\mathrm{tr}}^{-1} = 3K_{\mathrm{tr}}.$$

Proof. This is the Clebsch–Gordan decomposition for $\mathrm{Sym}^2(\mathrm{Sym}^2\mathbb{C}^2)$ [12]. The scalar summand has multiplicity one, so any invariant symmetric contact tensor is proportional to the displayed second-transvectant matrix. □

Benchmark Datum 5.2 (Majorana contact decomposition of the current benchmark)**.** The source-consistent Majorana matrix in the current benchmark is not a pure $(\mathbb{CP}^1, \mathcal{O}(2))$ Veronese matrix. It requires the unique orthogonal second-transvectant contact component

$$\frac{M_R}{M_*} = M_V + \zeta K_{\mathrm{tr}},$$

with

$$\zeta = 0.1076472949 + 0.0736514853\, i.$$

The contact fraction is

$$\frac{\|\zeta K_{\mathrm{tr}}\|}{\|M_R/M_*\|} = 1.304275 \times 10^{-1}.$$

The companion machine audit that produced the previous long draft decomposes the inverse-seesaw reconstructed M_R into the Veronese subspace plus the orthogonal trace/contact direction. The Veronese-plus-contact residual is at machine precision, whereas the Veronese-only residual equals the displayed contact fraction. This means the contact is not a negligible correction to the present PMNS-compatible branch. The lean note records the datum and the geometrically fixed tensor direction; the full matrices, basis cards, and norm conventions are deferred to companion artifacts.

Observation 5.3 (Negative scan for simple hidden-sector origins of ζ)**.** Within the currently tested simple hidden-quotient and small monomial clockwork classes, ζ is a conditional datum rather than a derived number.

The scan evidence is limited but sharp within that tested class. The contact-sensitivity scan gives the loose PMNS-compatible windows

$$\Delta s = 3.208798 \times 10^{-5}, \qquad \Delta\delta = 5.424119 \times 10^{-5} \text{ rad}.$$

Removing the contact spoils the PMNS and mass-splitting replay. A minimal hidden D-quotient leaves modulus and phase flat directions. A small real-coefficient monomial or clockwork phase with denominator $N \leq 6$ misses the target phase by at least

$$4.471595 \times 10^{-1} \text{ rad},$$

and the first root of unity inside the loose phase window appears only at

$$N = 178.$$

Thus no small hidden quotient or small monomial clockwork mechanism in the current evidence set predicts the complex value of ζ. A future non-minimal hidden sector may still do so, but it would be a new assumption that must be audited with its threshold and UV cost.

6 Optional Route B: Hidden Transvectant Origin for ζ

The negative scan above does not forbid a hidden origin for ζ. It only says that the currently tested small hidden quotients and small monomial clockwork classes do not predict it. This section is an optional conditional extension, not part of the Route-A theorem core. The conservative Route-B move is therefore to add a new hidden-sector theorem: derive the tensor direction from the unique transvectant K_{tr}, and isolate the remaining burden in a hidden phase/radial sector that supplies the complex number.

Theorem 6.1 (Post-$B-L$ sterile transvectant messenger)**.** *Let*

$$V = H^0(\mathbb{CP}^1, \mathcal{O}(2)).$$

After $B-L$ breaking, let

$$N_i = P_{\nu^c}(16_i), \qquad N \in V,$$

be the right-handed neutrino family vector selected by the Majorana source projector. Let

$$X \in V^\vee$$

be a visible-singlet sterile family messenger. At the post-$B-L$ matching scale, consider the coordinate-free superpotential

$$\frac{W_R}{M_*} = \frac{1}{2} M_V(N,N) + \lambda X(N) - \frac{1}{2} K_{\mathrm{tr}}^{-1}(X,X),$$

where

$$M_V \in \mathrm{Sym}^2 V^\vee, \qquad K_{\mathrm{tr}} \in \mathrm{Sym}^2 V^\vee, \qquad K_{\mathrm{tr}}^{-1} \in \mathrm{Sym}^2 V.$$

In a chosen basis this is

$$\frac{W_R}{M_*} = \frac{1}{2} N^T M_V N + \lambda X^T N - \frac{1}{2} X^T K_{\mathrm{tr}}^{-1} X,$$

where $K_{\rm tr}$ is the unique $SL(2)$-invariant contact direction from Lemma 4.1. Integrating out X gives

$$\frac{W_R^{\rm eff}}{M_*} = \frac{1}{2}\left(M_V + \lambda^2 K_{\rm tr}\right)(N,N).$$

Thus the benchmark contact is generated if

$$\lambda^2 = \zeta.$$

Proof. The X-F-term is the equality in V

$$\frac{\partial W_R}{\partial X} = M_*\left(\lambda N - K_{\rm tr}^{-1} X\right) = 0.$$

Here $K_{\rm tr}^{-1} : V^\vee \to V$ is the inverse map induced by the non-degenerate bilinear form $K_{\rm tr}$. Thus

$$X = \lambda K_{\rm tr} N,$$

where $K_{\rm tr} : V \to V^\vee$. Substitution gives, in any basis,

$$\begin{aligned}\frac{W_R^{\rm eff}}{M_*} &= \frac{1}{2}M_V(N,N) + \lambda X(N) - \frac{1}{2}K_{\rm tr}^{-1}(X,X)\\ &= \frac{1}{2}M_V(N,N) + \lambda^2 K_{\rm tr}(N,N) - \frac{1}{2}\lambda^2 K_{\rm tr}(N,N)\\ &= \frac{1}{2}\left(M_V + \lambda^2 K_{\rm tr}\right)(N,N).\end{aligned}$$

Equivalently, in matrix notation,

$$\frac{W_R^{\rm eff}}{M_*} = \frac{1}{2}N^T M_V N + \frac{1}{2}\lambda^2 N^T K_{\rm tr} N.$$

This is exactly the displayed effective Majorana matrix. □

Remark 6.2 (Scope of the projector). The field $N_i = P_{\nu^c}(16_i)$ is a post-$B-L$-breaking EFT variable. Before symmetry breaking, a Spin(10)-singlet messenger cannot select ν^c from the 16_i without an additional Majorana source or projector. Route B therefore assumes such a post-breaking projector, for example from a $(B-L) = -2$ Majorana source sector, rather than deriving it from face projection alone. In this lean note P_{ν^c} is therefore an input of the post-$B-L$ EFT matching problem, not a result of the face-projection principle.

Benchmark Datum 6.3 (Target coefficient and square-root coupling)**.** For the current benchmark,

$$\zeta_{\rm tar} = 0.1076472949 + 0.0736514853\, i.$$

Its polar data are

$$|\zeta_{\rm tar}| = 0.13043190325293763, \qquad \arg\zeta_{\rm tar} = 0.600038020318215.$$

One square-root coupling is

$$\lambda = \sqrt{\zeta_{\rm tar}} = 0.34502115743308964 + 0.10673473744038917\, i,$$

with

$$|\lambda| = 0.3611535729477664.$$

This is a moderate hidden coupling; it is not a high-power clockwork suppression.

Observation 6.4 ($\mathbb{Z}_{178}$ phase diagnostic)**.** A discrete phase can approximate the benchmark phase by

$$\zeta_{178} = |\zeta_{\rm tar}| \exp\left(\frac{2\pi i\, 17}{178}\right).$$

Since

$$\frac{2\pi\, 17}{178} = 0.6000794956295111,$$

the phase error is

$$4.1475311296 \times 10^{-5} \text{ rad},$$

which lies inside the loose phase window

$$5.424119 \times 10^{-5} \text{ rad}.$$

The direct complex error is

$$|\zeta_{178} - \zeta_{\rm tar}| = 5.4097037899 \times 10^{-6}.$$

This does not make ζ an exact prediction. It supplies a precise discrete-phase candidate; exact equality would require either a small axion shift or a continuous hidden phase. The Route-B hidden-messenger theorem uses only $\lambda^2 = \zeta$; the $\mathbb{Z}_{178}$ value is kept as a side diagnostic, not as a premise of the benchmark branch.

Observation 6.5 (Nonperturbative hidden-spurion option)**.** An alternative is

$$\zeta = Ae^{-S_H + i\theta_H}.$$

For $A = 1$,

$$S_H = -\log|\zeta_{\rm tar}| = 2.0369040025655094, \qquad \theta_H = \arg\zeta_{\rm tar}.$$

This is naturally interpreted as a nonperturbative hidden-sector spurion. It is a conditional physical origin, not a weak-coupling derivation from face-projection alone.

Lemma 6.6 (Minimal post-$B-L$ R-selection rule)**.** *At the post-$B-L$ matching scale, take the superpotential to have $R(W) = 2$. A minimal selection rule that keeps the sterile messenger Majorana-only is*

object	*role*	$U(1)_R$
16_i *components*	*matter multiplets, including* N_i	1
$H, T, \bar{T}$	*Higgs doublet/triplet fields*	0
$X \in V^\vee$	*sterile transvectant messenger*	1
$\lambda,\ M_V,\ K_{\rm tr}^{-1}$	*matching tensors/spurions*	0

together with the post-breaking projector

$$N_i = P_{\nu^c}(16_i).$$

Then the Route-B Majorana terms

$$X(N), \qquad K_{\rm tr}^{-1}(X, X), \qquad M_V(N, N)$$

all have $R = 2$, while ordinary Dirac and triplet Yukawa terms

$$16_i 16_j H, \qquad 16_i 16_j T$$

also have $R = 2$. However any superpotential operator with at least one additional X inserted into a Dirac or colored-triplet operator has

$$R = 2 + m, \qquad m \geq 1,$$

and is therefore forbidden if the R-selection rule is exact at the matching scale.

Proof. The allowed Route-B terms have

$$R(X) + R(N) = 1 + 1 = 2, \qquad 2R(X) = 2, \qquad 2R(N) = 2.$$

The baseline Dirac and triplet Yukawa terms have

$$R(16_i) + R(16_j) + R(H \text{ or } T) = 1 + 1 + 0 = 2.$$

Adding $m \geq 1$ sterile messengers to such an operator shifts the superpotential charge to $2 + m$, so the operator is not allowed. This proves the selection rule at the holomorphic matching level. If later R-breaking spurions are introduced, they must be charged so that these forbidden operators remain absent to the order being audited. □

Remark 6.7 (Status of the R-selection rule). The $U(1)_R$ assignment in Lemma 6.6 is used here as a holomorphic EFT matching selection rule. It is not yet claimed to be an anomaly-free microscopic symmetry. A UV implementation would require either an anomaly-free continuous or discrete R-symmetry, or a spurionic origin whose R-breaking insertions do not regenerate X-decorated Dirac or colored-triplet operators within the audited order.

Theorem 6.8 (Tree-level canonical matching silence of Majorana-only hidden contact)**.** *In a canonical-normalized post-$B - L$ EFT at the matching scale, use Route B as a post-$B - L$-breaking matching theorem, with*

$$N_i = P_{\nu^c}(16_i)$$

as above. Assume the hidden messenger sector obeys the R-selection rule of Lemma 6.6:

$$X \text{ is a visible singlet}, \qquad X \text{ couples only to the Majorana channel } K_{\mathrm{tr}}(N, N),$$

and X has no couplings to Dirac Yukawa operators or colored-triplet operators. If integrating out X gives

$$\Delta M_R / M_* = \zeta K_{\mathrm{tr}},$$

then the visible one-loop threshold vector, the Dirac Yukawa matrices, and the dimension-five colored-triplet Wilson tensors are unchanged relative to the benchmark.

Proof. Since X is a visible singlet,

$$\Delta b_1^X = \Delta b_2^X = \Delta b_3^X = 0,$$

so it contributes no non-universal visible one-loop threshold. The Majorana-only selection rule gives

$$\delta Y_u = \delta Y_d = \delta Y_e = \delta Y_{\nu D} = 0.$$

The same rule forbids hidden insertions in the colored-triplet Yukawa couplings and inverse triplet propagator, so

$$\delta C_{5L} = \delta C_{5R} = 0.$$

If ζ equals the benchmark value, then M_R is the benchmark Majorana matrix and the type-I seesaw replay is unchanged. Thus the hidden contact can generate the missing Majorana tensor without moving the visible flavor, proton-decay, or one-loop threshold audits. □

Observation 6.9 (Canonical basis caveat)**.** The previous theorem is a tree-level matching statement in a canonical-normalized EFT basis. A microscopic completion with a finite messenger interval can generate a right-handed-neutrino wave-function correction of order

$$\delta Z_N \sim \frac{|\lambda|^2}{16\pi^2} \log \frac{M_*}{M_X}.$$

For the benchmark value,

$$\frac{|\lambda|^2}{16\pi^2} = 8.259697 \times 10^{-4}.$$

This is small, but not automatically smaller than the narrow contact-sensitivity windows. Therefore the benchmark matrices in this lean note are defined after canonical normalization. If a microscopic model predicts a non-negligible δZ_N, the companion seesaw replay must use

$$Y_{\nu D} \mapsto Y_{\nu D}\left(1 - \frac{1}{2}\delta Z_N\right),$$

$$M_R \mapsto M_R - \frac{1}{2}\left(\delta Z_N^T M_R + M_R \delta Z_N\right),$$

before comparing to the PMNS benchmark.

The compact numerical checks for this optional Route-B extension are recorded separately in Appendix B; they are algebraic matching checks, not companion phenomenology audits.

7 Theorem Boundary

Theorem 7.1 (Conditional benchmark boundary)**.** *Given the face-projection data, the $O(2)$ index data, the companion benchmark audits, and either a benchmark-specified Majorana datum $\mathcal{D}_\zeta$ or a Route-B hidden messenger producing $\mathcal{D}_{\text{hidden zeta}}$, the current construction defines one internally checked conditional EFT branch:*

$$\frac{\mathcal{D}_{\text{face}} + \mathcal{D}_{\text{index}} + (\mathcal{D}_\zeta \text{ or } \mathcal{D}_{\text{hidden zeta}}) + \mathcal{D}_{\text{companion audits}}}{\rightsquigarrow \text{ one conditional benchmark branch.}}$$

This does not support the stronger implication

$$\textit{PSLT/Another Physics alone} \Longrightarrow \textit{a unique GUT with predicted } \zeta.$$

Proof. A complete microscopic GUT requires an explicit gauge group, chiral matter representations, anomaly cancellation, a breaking potential, Yukawa and Majorana sectors, baryon-violating operators, and threshold matching. Face projection motivates the Spin(10) representation geometry. The $O(2)$ index protects the family count. Neither supplies all action-level data. The remaining branch data A1–A7 and the benchmark-level checks are conditional inputs or companion artifacts, and they are required whether ζ is supplied as a benchmark datum or generated through Route B. The Route-B messenger derives ζK_{tr} only after adding a hidden sterile messenger, phase/radial data, and Majorana-only selection rules. It therefore improves the conditional EFT explanation of ζ, but it is still not a PSLT-only prediction and it does not replace the companion phenomenological audits. □

Corollary 7.2 (Status of the rewritten theory)**.** *The complete claim of this lean version is*

$$\boxed{\begin{array}{c}\textit{face-projection motivation} + \textit{ conditional } \text{Spin}(10) \textit{ EFT} \\ \neq \textit{unconditional PSLT-only GUT proof.}\end{array}}$$

The rewrite is nevertheless useful: it gives a coherent representation-level reason why Spin(10) *is the natural minimal six-face family object and why three families are protected by an independent $O(2)$ index.*

8 Minimal Reproducibility Manifest

The lean note has a small, explicit reproducibility surface. It does not include the deferred full flavor/proton/threshold scans.

item	artifact or command
paper source	`paper/gut_framework.tex`, `paper/refs.bib`
compiled note	`paper/gut_framework.pdf`
D5 appendix check	`python3 code/verify_d5_half_spin_hypercube.py`
D5 outputs	D5 summary, weights, and report in `output/d5_half_spin/`
Route-B algebra check	`python3 code/verify_hidden_zeta_origin.py`
Route-B output	verification JSON in `output/hidden_zeta_origin/`
status log	`roadmap.md`

The reproducible numerical claims inside this TeX draft are therefore only the D5 half-spin branching check, the hypercharge normalization check, and the Route-B algebraic ζK_{tr} matching check. All publication-grade phenomenology remains outside this manifest until the companion audits are restored.

9 Deferred Calculations

The previous version contained detailed scan and audit material for Yukawa textures, $d = 5$ proton decay, threshold corrections, source sectors, clockwork rescue, no-web provenance, and theorem ledgers. Those calculations are not deleted from the project history: they are preserved in the backup made before this rewrite and in the corresponding output artifacts. They are omitted from this TeX draft because their status relative to the face-projection principle is not yet settled. This lean note defines the representation-level and index-level skeleton; benchmark-level numerical checks are deferred to companion artifacts. The only numerical Route-B checks kept in this draft are the algebraic ζ, K_{tr}, and phase-spurion checks needed to support the hidden messenger theorem.

The conservative paper strategy is Route A as the main note: keep the proved representation and index skeleton in the foreground, and treat Route B as a candidate hidden-sector mechanism or companion-paper seed. Promoting Route B to the main line would require restoring only those flavor, proton-decay, threshold, and source-sector calculations that survive the R-selection rule and canonical-normalization replay. In short,

Route A	main paper: conditional face-projection model note,
Route B	candidate mechanism or companion: hidden origin for ζ.

10 Lean-Note Status Ledger

The logical status of this version is deliberately narrower than a complete GUT construction:

status	content
proved/constructed	Spin(10) face skeleton; $\mathrm{Spin}^c(\mathbb{CP}^1, \mathcal{O}(2))$ index; K_{tr} uniqueness; optional Route-B Schur complement.
conditional inputs	$C = \mathbb{CP}^1$, $R = p_+ + p_-$; protected Spin^c carrier; no unpaired SM-charged exotics; P_{ν^c}; R-selection rule; hidden phase/radial sector.
deferred audits	full flavor fit; $d = 5$ proton-decay Wilson tensors; threshold/proton companion audits; source-sector UV origin.
not claimed	PSLT-only unique GUT proof.

11 Conclusion

The clean result of the rewrite is a bounded theorem stack. A Spin(10) half-spinor supplies a standard minimal six-face object; the $\mathrm{Spin}^c(\mathbb{CP}^1, \mathcal{O}(2))$ carrier supplies three protected copies; and the second-transvectant singlet supplies the unique Majorana contact direction. These are mathematical inputs and consequences inside a conditional EFT branch. They do not by themselves determine the microscopic source sector, the complex hidden coefficient ζ, the full flavor fit, or the proton-decay Wilson tensors. Keeping this boundary explicit is the point of the lean Route-A version.

A D_5 Half-Spin Weight Hypercube

This appendix records the compact representation-theoretic calculation behind the six-face statement. It is not a new uniqueness theorem; it is the weight-level explanation of the standard Spin(10) half-spinor branch used in the main text [4, 12].

Let $e_1, \ldots, e_5$ be an orthonormal basis for the D_5 weight space. The roots of D_5 are

$$\pm e_a \pm e_b, \qquad a \neq b.$$

Choose the positive-chirality half-spin representation to have weights

$$\Lambda_{16} = \left\{ \frac{1}{2} \sum_{a=1}^{5} s_a e_a : s_a = \pm 1, \quad \#\{a : s_a = -1\} \equiv 0 \pmod 2 \right\}.$$

There are $2^{5-1} = 16$ such weights. The Pati–Salam face is the split

$$D_5 \supset D_3 \oplus D_2, \qquad \mathrm{Spin}(6) \times \mathrm{Spin}(4) \simeq \mathrm{SU}(4)_C \times \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R.$$

Use s_1, s_2, s_3 for the D_3 spinor weights and s_4, s_5 for the D_2 spinor weights. Define

$$m_C = \#\{a \leq 3 : s_a = -1\}, \qquad m_W = \#\{a = 4, 5 : s_a = -1\}.$$

The chirality condition says $m_C + m_W$ is even, so m_C and m_W have the same parity. Therefore the weights split into exactly two blocks:

m_C parity	m_W parity	number of weights	Pati–Salam block
even	even	$4 \cdot 2 = 8$	$(4, 2, 1)$
odd	odd	$4 \cdot 2 = 8$	$(\bar{4}, 1, 2)$.

Thus

$$16 \to (4, 2, 1) \oplus (\bar{4}, 1, 2).$$

One convenient convention for the $\mathrm{SU}(4)_C$ component is

first three signs	$\mathrm{SU}(3)_C$	$B - L$
zero minus signs	1	-1
two minus signs	3	$1/3$
one minus sign	$\bar{3}$	$-1/3$
three minus signs	1	1.

For the weak D_2 signs, take

$$(s_4, s_5) = (+, +), (-, -) \quad \Longrightarrow \quad \mathrm{SU}(2)_L \text{ doublet}, \quad T_{3L} = \frac{s_4 + s_5}{4}, \quad T_{3R} = 0,$$

and

$$(s_4, s_5) = (+,-), (-,+) \quad \Longrightarrow \quad \mathrm{SU}(2)_R \text{ doublet}, \quad T_{3R} = \frac{s_4 - s_5}{4}, \quad T_{3L} = 0.$$

Applying

$$Y = T_{3R} + \frac{B-L}{2}$$

then gives the six visible Standard-Model faces:

field	$\mathrm{SU}(3)_C \times \mathrm{SU}(2)_L$	Y	multiplicity
Q	$(3,2)$	$1/6$	6
L	$(1,2)$	$-1/2$	2
u^c	$(\bar{3},1)$	$-2/3$	3
d^c	$(\bar{3},1)$	$1/3$	3
ν^c	$(1,1)$	0	1
e^c	$(1,1)$	1	1.

The multiplicities sum to 16, so this is a complete half-spinor, not a collection of separately chosen low-energy fragments.

The appendix calculation is reproduced by

`code/verify_d5_half_spin_hypercube.py`.

The generated summary reports

$$\#\Lambda_{16} = 16, \qquad (4,2,1) : 8, \qquad (\bar{4},1,2) : 8,$$

the six field multiplicities

$$(Q, L, u^c, d^c, \nu^c, e^c) = (6,2,3,3,1,1),$$

and the same normalization check as in the main text,

$$\mathrm{Tr}\, Y^2 = \frac{10}{3}, \qquad \mathrm{Tr}\, T_{3L}^2 = 2, \qquad k_Y = \frac{5}{3}.$$

B Optional Route-B Algebraic Checks

This table checks only the optional Route-B Schur-complement algebra. It is not a substitute for the deferred full flavor, proton-decay, threshold, or source-sector audits.

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check	value	role
$\lvert\zeta_{\rm tar}\rvert$	0.13043190325293763	target modulus
$\arg\zeta_{\rm tar}$	0.600038020318215	target phase
$\sqrt{\zeta_{\rm tar}}$	$0.345021 + 0.106735\,i$	hidden messenger coupling
$\lvert\lambda\rvert$	0.3611535729477664	moderate coupling
$\lVert\lambda^2 K_{\rm tr} - \zeta K_{\rm tr}\rVert_F$	1.20185×10^{-17}	Schur-complement match
$\lVert K_{\rm tr}(3K_{\rm tr}) - I\rVert_F$	3.84593×10^{-16}	inverse convention
$2\pi(17/178) - \arg\zeta$	4.14753×10^{-5} rad	side phase diagnostic
$\lvert\zeta_{178} - \zeta_{\rm tar}\rvert$	5.40970×10^{-6}	discrete approximation
$-\log\lvert\zeta\rvert$	2.0369040025655094	instanton-spurion option
$\lvert\lambda\rvert^2/(16\pi^2)$	8.25970×10^{-4}	canonical-basis audit scale
$\Delta b^X_{\rm vis}$	$(0, 0, 0)$	visible threshold silence

Table 1: Optional Route-B algebraic verification.

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